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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT		
S. No.	**University ID**	**Name**
1	2420090110	Manikonda Pranavi
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Name of the Guide		Anugu Swapna

Abstract

The Adaptive Campus Canteen Pre-Ordering and Smart Pickup System is a web-based application designed to improve the efficiency of food ordering and pickup in college canteens. During break hours, students often experience long queues, overcrowding, and delays in receiving their meals, leading to inconvenience and loss of valuable time. The proposed system enables students to browse the available menu, place food orders in advance, select a preferred pickup time, and receive a digital token or QR code for quick food collection. This approach simplifies the ordering process, minimizes waiting time, and provides a more convenient dining experience for students.

A key feature of the proposed system is its adaptive demand management mechanism, which continuously monitors incoming orders and responds to changing demand patterns. When the number of orders for a particular food item exceeds a predefined threshold, the system automatically notifies canteen staff to prepare additional food in advance. It also updates food availability and pickup schedules based on real-time order conditions, helping staff manage resources more effectively. By adapting to changing order volumes, the system reduces the chances of food shortages, unnecessary delays, and food wastage while ensuring efficient canteen operations.

The application provides separate modules for students, canteen staff, and administrators. Students can register, log in, browse the menu, place and track orders, and collect food using a digital token or QR code. Canteen staff can manage menu items, monitor incoming orders, update order status, and respond to adaptive alerts generated by the system. Administrators can manage users, monitor daily transactions, generate reports, and analyze order trends to improve overall canteen performance. These integrated modules ensure smooth communication and efficient coordination among all stakeholders.

The proposed system will be developed using React.js for the frontend, Node.js and Express.js for the backend, and MySQL for database management. Git and GitHub will be used for version control, while Jira will support Agile project management through sprint planning and task tracking. By following the principles of Adaptive Software Engineering, the system dynamically responds to changing user requirements and operational conditions, making it a practical, scalable, and cost-effective solution for enhancing campus canteen services while improving user satisfaction and operational efficiency.

Signature of the Guide HOD